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RESEARCH ARTICLE

3D Segmentation of Necrotic Lung Lesions in CT Images Using Self-Supervised Contrastive Learning

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This work involved human subjects or animals in its research. The authors confirm that all human/animal subject research procedures and protocols are exempt from review board approval.

ABSTRACT Deep convolutional neural networks (CNN) are often trained on 2D annotations created by radiologists following RECIST guidelines to segment lesions in 3D medical images. Three-dimensional segmentation is conducted by segmenting each lesion slice-by-slice on the axial direction and stacking the 2D segmentation masks into 3D. However, the performance of such models is inherently biased by the appearance of most of the lesions in the training dataset. Herein we propose an approach to generate accurate 3D segmentations of underrepresented necrotic lung lesions. Our proposed approach applies two novel augmentation techniques for contrastive learning pretraining: dependency augmentation that captures inter-slice dependencies, and distance transform-based mask-out augmentation imitating necrotic lesions. In dependency augmentation, cosine similarity within RECIST bounding box is applied to construct positive pairs from 2D image slices of the same lesion in the current 3D volume and across longitudinal scans. We further compared contrastive learning architectures, Momentum Contrast (MoCo) and Bootstrap Your Own Latent (BYOL), based upon two internal 3D testing sets, one with regular lung lesions and the other with necrotic lung lesions, and a public 3D DeepLesion lung lesion testing set. MoCo with both proposed augmentations demonstrated the best performance among all methods that were compared. Specifically, it 1) improved Dice similarity coefficient (DSC) by 8.42% over baseline model trained from scratch and 2.40% over ImageNet pretrained model on the 3D necrotic lung lesion set; 2) achieved better segmentation performance on necrotic lesions with 10% of labeled data for supervised fine-tuning compared with the baseline model trained with all labels from scratch.

INDEX TERMS Deep learning, necrotic lung lesion, lesion segmentation, self-supervised learning, semantic segmentation.

I. INTRODUCTION

In oncology clinical trials, efficacy is assessed by measuring the change in tumor burden over time, using the Response Evaluation Criteria in Solid Tumors (RECIST) [1]. Prior to treatment, radiologists find all malignant lesions, choose a subset to follow qualitatively, called “target” lesions, and

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make unidimensional size measurements (typically manually drawing an outline of the lesion boundary on a single slice). They also document all other lesions, which are followed qualitatively as “non-target” lesions. At each on-treatment assessment timepoint, reviewers measure the target lesions, compare their aggregate size to prior timepoints, assess the non-target lesions for total disappearance or unequivocal growth, and search for new lesions, then combine these assessments into an “overall response” for that timepoint.

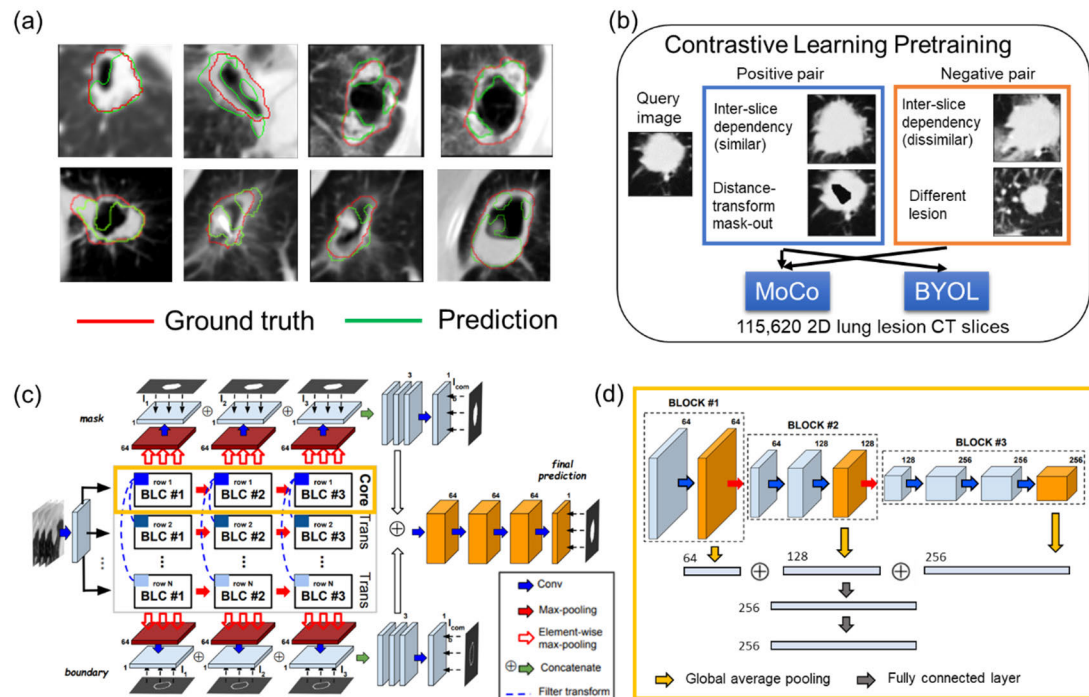


FIGURE 1. (a) Example bad segmentation results on necrotic lung lesions. (b) Contrastive learning with dependency augmentation that captures inter-slice dependency and distance transform-based mask-out augmentation. (c) SIBA-Net architecture. (d) Contrastive pretraining architecture derived from the core branch of SIBA-Net.

The sequence of overall responses for all timepoints during treatment is then used to determine endpoints such as progression-free survival and objective response rate.

RECIST-based endpoints are strong predictors of clinical outcome in large cohorts, but do not correlate well with survival in small cohorts, so there is strong interest and active research on better imaging-based methods of evaluating efficacy to replace RECIST. For these more sophisticated approaches, such as extracting radiomic features to assess changes in the tumor microenvironment [2] and modeling of lesion growth kinetics [3], unidimensional measurements of lesion size are insufficient, and accurate 3D segmentation of lesions in CT images becomes crucial.

Recently, deep convolutional neural networks (CNNs) trained on 2D RECIST image delineations have been applied to segment lesions in a slice-by-slice manner [4], [5], where 2D slices are stacked to generate 3D segmentations, thereby significantly reducing the labor consumption for 3D manual delineation. RECIST analysis done in the course of clinical trials is useful for 3D segmentation of lesions because it leverages the work already done by human experts to locate lesions in 3D CT scans, and thus provide the starting point for the 3D segmentation that would enable methods such as radiomics and lesions growth kinetics to eventually supersede RECIST. However, the performance of these segmentation models often suffers on lung lesions that show necrosis [6], [7] as illustrated in Fig. 1a. These lesions contain dying tumor tissue in response to treatment, leading to hypodense cores. Since the training of the 2D CNN-based segmentation models is often biased towards the majority of lesions

showing regular appearance in the training dataset, the trained models tend to exclude the hypodense necrotic regions from the segmentation.

Existing approaches for handling data imbalance issue include resampling of under- and over-represented classes in supervised learning [8], [9] and resampling of samples from hard tail classes in contrastive learning [10]. Self-supervised contrastive learning leverages large unlabeled datasets by contrasting augmented positive and/or negative image pairs to learn robust features that are closely relevant to the image domain, and the pretrained models can be tuned effectively with limited labels [11], [12], [13], [14]. Furthermore, contrastive learning was found to be more robust in handling under-represented classes than supervised ImageNet-based pretraining [15].

Under a contrastive learning framework, three-dimensional medical images and videos contain spatial and temporal continuity that could be learned as invariance and embedded the learning process. In 3D CT and MR images, Zeng et al. [16] used the spatial continuity of lesions to generate positive pairs of neighboring slices and tracked lesions across multiple timepoints. Similar to the slice-wise transition in medical images, adjacent frames in videos could carry useful continuity information. Feichtenhofer et al. [17] selected multiple video clips within a one-minute timespan as positive pairs and found improvements across various contrastive learning frameworks and downstream tasks. Since objects could have substantial difference across spatial and temporal dimensions, the spatial distance and time span may not be the best for positive pair selection. Qian et al. [18] selected

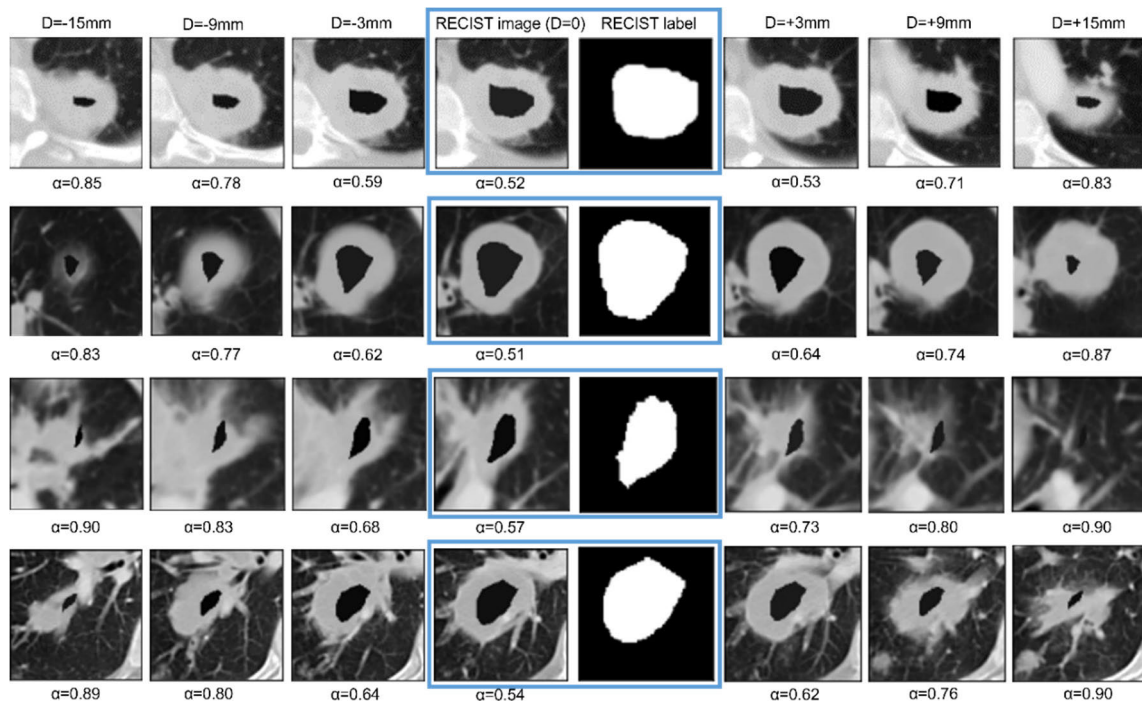


FIGURE 2. Example distance transform-based mask-out in 7 z-slices of four lesion volumes. D is the distance from the current image to the RECIST image on the axial direction. α is the distance transform threshold computed as a function of D using Equation (3). The central area of the CT image that has distance transform value $>\alpha$ are masked out. As the images approach the upper and lower end of the lesion, α gets larger, and the masked-out region is smaller.

two clips from the same video to generate positive pairs with closer clips having higher probabilities to be selected as positive pairs. Han et al. [19] used optical flow to help select positive pairs among video clips. Peng et al. [20] proposed self-paced contrastive learning where the network started with confident positive pairs and gradually learned from less confident ones, using confidence for positive pair selection and adaptively change the confidence threshold as the model converges. But self-paced learning is harder to converge and selecting a good learning pace was challenging.

Artifacts have been introduced in the construction of positive pairs to improve the model's robustness in artifact cases during contrastive learning. Xu et al. [21] perturbed the input and feature space to simulate shadow artifacts in prostate ultrasound images to learn consistent feature representations between normal and perturbed images. Neto et al. [22] generated human face images with face masks to learn semantic features robust to the presence of masks and therefore improved the performance of face recognition.

Herein we propose a novel approach for the robust and accurate segmentation of necrotic lung lesions using contrastive learning. Given the abundance of unlabeled 2D CT image slices and the relatively limited number of labeled 2D RECIST images, we introduce a framework conducting self-supervised contrastive learning based on RECIST delineations and unlabeled images, followed by supervised fine-tuning using 2D RECIST images and delineations. Contrastive learning relies on strong augmentation between

positive pairs for the deep learning models to learn invariant features in the images. For the pretraining, in addition to the standard data augmentation methods, we utilize dependency augmentation that captures inter-slice dependencies: the slices of the lesion from z-dimension and longitudinal scans at different timepoints are selected using cosine similarity criteria to augment for positive pairs. This augmentation creates strong and natural variations to the positive pair and enables the encoder to learn the robust, invariant, and dependent features describing the lesion and its surrounding region between slices and timepoints. In addition, based on the RECIST delineations, we introduce a distance transform-based mask-out augmentation to simulate necrotic lung lesions and include them in the contrastive learning. By pairing a regular-looking lesion and a necrotic lesion into a positive pair, the model learns the invariant boundary part of the lesion, and therefore the hypointense necrotic core would not be excluded. Experiments were carried out on two private testing datasets, one for necrotic lung lesions and one for lesions with regular appearance, and one public testing dataset [23] of randomly selected lung lesions, all manually annotated in 3D by experts. We evaluated the proposed approach through a comprehensive ablation study, comparing different augmentation methods under two contrastive learning frameworks, MoCo [12], [13] and BYOL [14]. We also compared our proposed approach with two state-of-the-art CNN-based segmentation models, HRNet [24] and nnU-Net [25]. HRNet learns strong

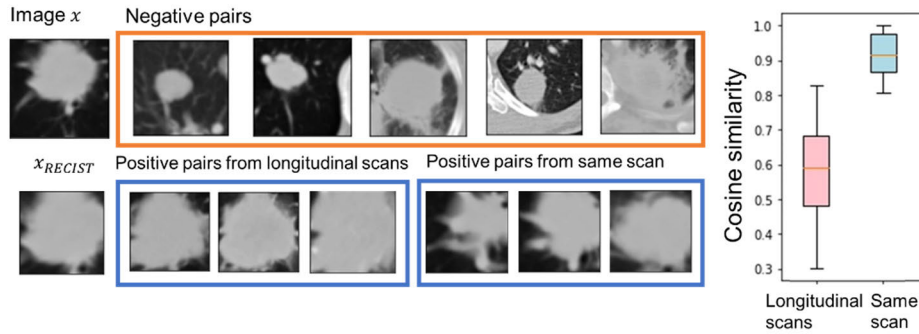


FIGURE 3. Example negative pairs of image x are in the orange box. Candidates for inter-slice positive pairs (from longitudinal CT scans and from the same CT scan) of resized cropped x_{RECIST} are in the blue boxes. The boxplots of cosine similarity in all candidate positive pairs for x_{RECIST} are on the right, with pink for longitudinal scans and blue for same CT scan. The cosine similarity values in positive pairs from the same CT scan are higher than the positive pairs from longitudinal scans, indicating the lesion has more variation between longitudinal scans than the z-direction variation in a particular scan. Overall, cosine similarity is a robust and quantifiable metric for positive pair construction.

high-resolution representations by connecting high-and-low resolution convolutions in parallel, where there are repeated multi-scale fusions across parallel convolutions [26]. HRNet has demonstrated great segmentation and detection performance in real-world pictures such as Cityscapes and PASCAL dataset [24] as well as in medical images [27], [28], [29]. The nnU-Net utilizes the U-Net architecture with automatic configuration of preprocessing, network architecture, training and post-processing for any new task which known to be one of the state-of-the-art approaches for various medical image segmentation tasks. Finally, we demonstrated the potential benefit of reducing the demand for labeled data in the supervised fine-tuning step following the pretraining step. The results on the three test datasets showed that the proposed approach led to a substantial improvement in segmentation accuracy on 3D necrotic lung lesions while maintaining performance on other lung lesions with regular appearance. As far as we know, there is no previous study focusing on using contrastive learning approach to improve the segmentation of necrotic lung lesions.

II. IMAGE ANALYSIS METHODS

A. PREPROCESSING

The lesion images were cropped to sizes of $3r \times 3r \times 2r$ in x , y , and z dimensions with the RECIST annotation at the center, where r is the radius of the lesion. We introduced random shifting in x - and y -dimensions during the cropping process to prevent centering effect of the lesion which could bias the model. Positional bias occurs when an object consistently appears in the same location in the images, e.g. the center, causing the deep learning model to learn and predict based on that specific location. To mitigate the bias, random shifting is employed in the training images to train the model robustly, such that it would be able to accurately segment the object of interest regardless of its location [30]. The 3D lesion volumes were resampled to resolution of $0.75 \text{ mm} \times 0.75 \text{ mm} \times 0.75 \text{ mm}$, and then each 2D slice was resize to 112×112 pixels.

B. MODEL FRAMEWORKS

We assess two self-supervised contrastive learning frameworks, MoCo [12], [13] and BYOL [14], with a VGG-16 backbone from the SiBA-Net architecture [5] as shown in Fig. 1b-d. Under our MoCo framework, in addition to the queue to store the embeddings of images in the original MoCo, we include two additional data fields which are the lesion identifier and the resized cropped images for each element of the queue to determine positive and negative pairs online. The lesion identifier consists of the trial number, site number, patient number, and lesion ID, which together form a unique ID for each lesion. Resized cropped images x_{RECIST} are generated by cropping the lesion images x to the tight bounding boxes around the RECIST delineations, followed by resizing the cropped images to 112×112 pixels. Details of positive pair construction are in the next section. As multiple positive pairs exist for one query image, we modified the InfoNCE loss as shown in (1):

$$L = -\log \frac{\sum \exp(z_\theta \cdot z_\varepsilon^{k+} / \tau)}{\sum \exp(z_\theta \cdot z_\varepsilon^{k+} / \tau + z_\theta \cdot z_\varepsilon^{k-} / \tau)} \quad (1)$$

where z_θ is the query embedding, z_ε^{k+} and z_ε^{k-} are the embeddings from positive and negative pairs, respectively. $\tau = 0.05$, and $\varepsilon = m\varepsilon + (1 - m)\theta$ with $m = 0.999$.

For each input image in BYOL, we randomly select one positive pair online within the current mini-batch. We use original negative cosine similarity loss for BYOL as shown in (2).

$$L = -\log \frac{q_\theta(z_\theta) \cdot z_\varepsilon^{k+}}{\|q_\theta(z_\theta)\|_2 \cdot \|z_\varepsilon^{k+}\|_2} \quad (2)$$

C. AUGMENTATION METHODS

In addition to conventional data augmentation methods, e.g. rotation, random cropping, and intensity perturbation, we introduce two novel augmentation methods.

First, multiple positive pairs are identified for each slice using a rule that captures inter-slice dependency: for a given

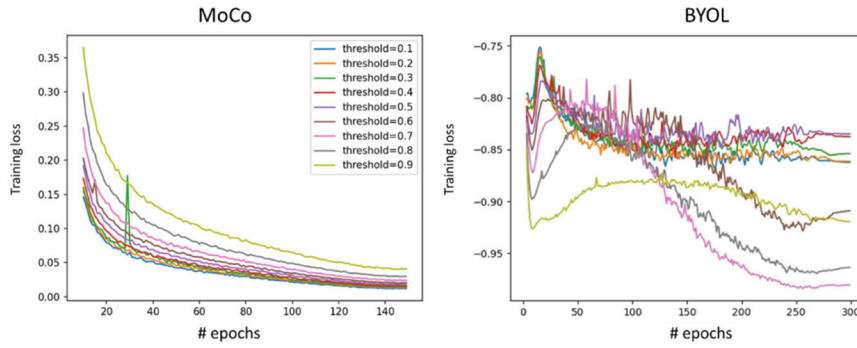


FIGURE 4. Training curve of contrastive learning with cosine similarity threshold 0.1 - 0.9 for MoCo and BYOL.

TABLE 1. Lesion segmentation performance for MoCo and BYOL with cosine similarity threshold of 0.1-0.9 on an independent 2D necrotic evaluation set for selection of optimal cosine similarity threshold.

		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
DSC	MoCo	0.7140 (0.1896)	0.7161 (0.1882)	0.7131 (0.1785)	0.7270 (0.1722)	0.7283 (0.1791)	0.7210 (0.1752)	0.7277 (0.1662)	0.7400 (0.1695)	0.7432 (0.1419)
	BYOL	0.7112 (0.1920)	0.7122 (0.1832)	0.7275 (0.1933)	0.7266 (0.1796)	0.7184 (0.1841)	0.7199 (0.1911)	0.7436 (0.1603)	0.7350 (0.1764)	0.7332 (0.1803)
HD	MoCo	14.12 (8.46)	13.48 (7.79)	14.12 (7.92)	13.49 (7.39)	12.58 (7.15)	13.57 (7.45)	13.58 (6.81)	12.82 (6.53)	12.45 (6.38)
	BYOL	15.09 (9.45)	14.48 (10.01)	14.33 (11.66)	14.29 (10.05)	13.62 (9.83)	13.39 (8.33)	13.65 (9.66)	13.08 (9.81)	13.47 (9.79)

TABLE 2. Ablation study to illustrate the impact of dependency augmentation that captures inter-slice dependency (Aug1) and distance transform-based mask-out (Aug2) on three test sets. Means are reported with standard deviations in the brackets. Best results on each dataset are in bold. * indicates statistical significance compared to results of MoCo with two augmentation methods.

	Aug1	Aug2	DSC			HD (mm)		
			Regular	DeepLesion	Necrotic	Regular	DeepLesion	Necrotic
Baseline			0.7534* (0.1404)	0.7037* (0.1872)	0.6276* (0.1739)	14.68* (6.35)	17.05* (6.90)	19.24* (8.86)
ImageNet			0.7780* (0.1406)	0.7810* (0.1232)	0.6878* (0.1510)	13.34* (7.34)	14.47* (7.97)	19.58* (10.57)
MoCo			0.7773* (0.1393)	0.7802* (0.1332)	0.6881* (0.1527)	13.34* (7.26)	15.34* (10.19)	20.69* (11.62)
	√		0.7837* (0.1407)	0.7834* (0.1153)	0.6968* (0.1437)	13.68* (7.39)	15.13* (9.37)	21.82* (12.93)
		√	0.7684* (0.1544)	0.7684* (0.1258)	0.6797* (0.1752)	13.46* (7.39)	14.85* (9.25)	14.37* (6.40)
	√	√	0.7840 (0.1418)	0.7950 (0.1235)	0.7118 (0.1489)	13.38 (7.30)	14.09 (9.12)	15.78 (8.09)
BYOL			0.7379* (0.1428)	0.6440* (0.2040)	0.6002* (0.1911)	16.07* (6.47)	21.60* (9.27)	20.41* (9.71)
	√		0.7718* (0.1396)	0.7672* (0.1237)	0.6454* (0.1616)	14.24* (7.51)	14.34* (7.82)	18.57* (9.26)
		√	0.7815* (0.1471)	0.7660* (0.1255)	0.6498* (0.1696)	12.29* (6.68)	14.05* (7.59)	15.59* (6.68)
	√	√	0.7701* (0.1473)	0.7754* (0.1172)	0.6823* (0.1515)	13.37* (7.84)	14.37* (8.16)	17.45* (8.97)

slice x , the slices from the same lesion on the current 3D scan and across all longitudinal scans are considered for the construction of positive pairs. Specifically, if the cosine similarity calculated by (3) between the two resized cropped images x_{RECIST}^1 and x_{RECIST}^2 is greater than a threshold T , the two images form a positive pair. Negative pairs are constructed with images from different lesions, as well as those from the

same lesion but the cosine similarity lower than the threshold.

$$\text{Cosine Similarity} = \frac{\langle x_{RECIST}^1, x_{RECIST}^2 \rangle_F}{\|x_{RECIST}^1\|_2 \cdot \|x_{RECIST}^2\|_2} \quad (3)$$

Second, given one unlabeled CT image and the corresponding RECIST delineation of the lesion, we perform distance transform inside the RECIST mask, normalize the value to

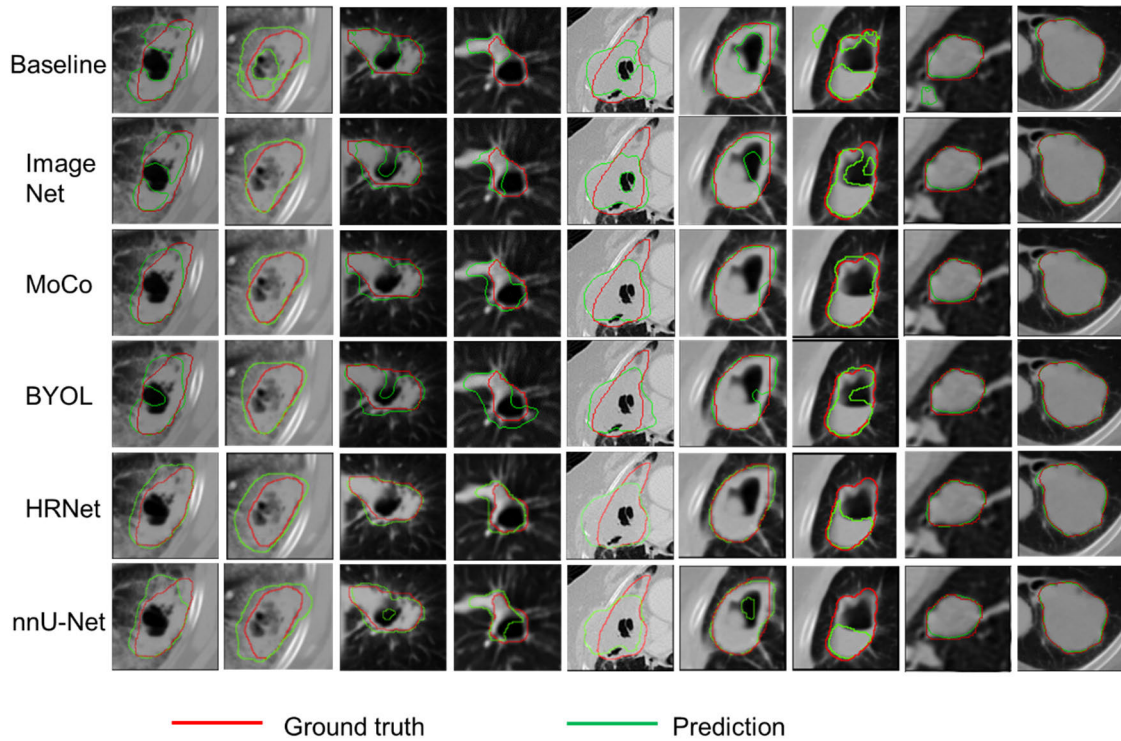


FIGURE 5. Visualization of segmentation result on examples of 7 necrotic (left) and 2 regular (right) lung lesions from Baseline model, ImageNet pretrained model, MoCo with two augmentation methods, BYOL with two augmentation methods, HRNet, and nnU-Net.

TABLE 3. Comparison with state-of-the-art segmentation models HRNet and nnU-Net on three test sets. Means are reported with standard deviations in the brackets. Best results on each dataset are in bold. * indicates statistical significance compared to results of MoCo with two augmentation methods.

	DSC			HD (mm)		
	Regular	DeepLesion	Necrotic	Regular	DeepLesion	Necrotic
HRNet[24]	0.7383* (0.1661)	0.7759* (0.1213)	0.6472* (0.1771)	13.18* (6.98)	13.84* (7.36)	17.41* (7.15)
nnU-Net[25]	0.7664* (0.1438)	0.7654* (0.1287)	0.6360* (0.1709)	13.01* (6.97)	16.01* (9.04)	18.30* (8.44)
MoCo+2Aug	0.7840 (0.1418)	0.7950 (0.1235)	0.7118 (0.1489)	13.38 (7.30)	14.09 (9.12)	15.78 (8.09)

the range of $[0, 1]$ with 0 at the boundary and 1 at the center of the lesion, and mask out the central area of the lesion on the CT image that has distance transform value higher than the threshold α . Threshold α is calculated by (4).

$$\alpha = \min[0.5 + \min\left(\frac{|D|}{r}, 0.4\right), 0.9] \quad (4)$$

where D is the distance between the current slice and the RECIST slice, and r is the radius of the lesion. α is kept in the range of $[0.5, 0.9]$. As the slices approach the upper and lower end of the lesion, α gets larger, and the masked-out region is smaller. This cropping-based augmentation is applied at a probability of $p = 0.6$. Example images of this augmentation are shown in Fig. 2.

III. EXPERIMENTAL METHODS

A. MATERIALS

Institutional Review Board approval was obtained from multiple sites, and informed consent was waived for this

retrospective study compliant with the Health Insurance Portability and Accountability Act. We collected 1,683 lung lesions from 1,121 participating subjects in multiple multi-center trials conducted during 2013-2021. All scans were deidentified and collected centrally for independent review. For each lung lesion, there are multiple timepoints, which results in 7,024 3D volumes, or 115,620 2D CT slices.

For each lesion, RECIST slice with 2D delineation by a radiologist and non-RECIST slices without delineation are utilized. For self-supervised pretraining, we used 115,620 CT slices from all lung lesions. For supervised fine tuning, 5,309 and 395 internal RECIST slices with 2D delineations were used for training and validation, respectively. We had three test sets: first was 250 internal 3D regular lung lesions, second was 140 3D lung lesions randomly selected from public DeepLesion dataset [23], and third was 52 internal 3D necrotic lung lesions. All internal test data were from different trials compared to training and validation dataset. Three imaging experts distributed the three test

sets and conducted annotations with 3D Slicer software independently.

B. PRETRAINING AND FINE-TUNING

We used a queue size of 30,000, mini-batch size of 300, and number of epochs of 150 for MoCo. We used mini-batch size of 200 and number of epochs of 300 for BYOL. Adam optimizer with initial learning rate of 10^{-3} and cosine learning rate decay was applied for both MoCo and BYOL. Besides the proposed augmentations, standard augmentations in contrastive learning including random cropping, brightness and contrast change, horizontal flipping, random rotation in (0° , 90° , 180° , 270°), and addition of random noise were used for all MoCo and BYOL models. For supervised fine tuning, we fine-tuned the entire CNN without freezing any layers. We used mini-batch size of 48, number of epochs of 100, and Adam optimizer with learning rate of 10^{-4} . We used random cropping for data augmentation in supervised training. Early stopping was applied once the validation loss stopped decreasing for 10 epochs. No post-processing was performed.

C. STATISTICAL ANALYSIS

Segmentation performance was assessed using Dice similarity coefficient (DSC) [31] and Hausdorff distance (HD) [32], which are metrics used regularly for assessing the accuracy of automatic segmentations. DSC focuses on the volume overlap, the value varies between 0 and 1, with 1 being perfectly overlapping. HD focuses on assessing the largest distance between the segmentation surfaces which shows the distance at the region with the worst segmentation quality.

For each test set, we repeated the supervised tuning 3 times to evaluate consistency at model level. Differences between our proposed MoCo model with two augmentation methods versus other models were assessed using paired *t*-test with significance level set as $p < 0.05$.

IV. RESULTS

A. VALIDITY OF COSINE SIMILARITY

The validity of cosine similarity can be justified from three aspects: computational robustness, computational efficiency, and capability to describe image-level similarities. Firstly, it is robust to different CT scanners and contrast phases due to the intensity normalization process. Secondly, computational efficiency comes with torch GPU computation. Thirdly, we show example candidate positive pairs and negative pairs for one image along with the cosine similarities in Fig. 3. The positive pairs are determined by two criteria: 1) the two lesion images are from the same lesion across longitudinal scans in any z-slices; 2) the RECIST cropped image have cosine similarity greater than threshold T . Negative pairs are images from different lesions. The lesion image in temporal positive pairs have deformation due to treatment response. In inter-slice positive pairs, the lesions demonstrated natural shape changes along the z-direction. Since the shape

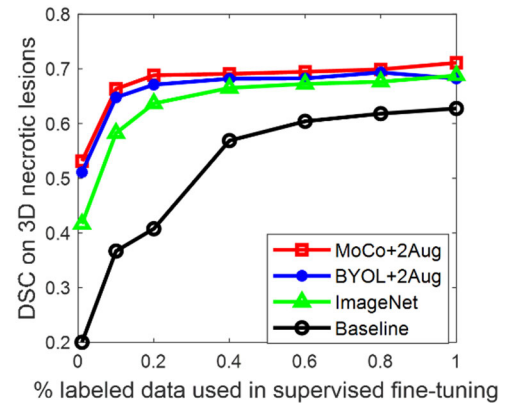


FIGURE 6. Comparison of DSC on 3D necrotic testing set between Baseline, ImageNet pretraining, MoCo with two augmentations, and BYOL with two augmentations. We vary the number of labels at 1%, 10%, 20%, 40%, 60%, 80%, and 100%.

changes in time domain and z-direction varies from lesion to lesion, cosine similarity is a more robust metric to capture the similarity compared to the time interval in [17] and spatial distances in [16]. Besides, using the RECIST cropped image allows the pretrained network to focus on the lesions rather than the surroundings, which provides an additional layer of guidance.

B. OPTIMIZATION FOR COSINE SIMILARITY THRESHOLD

MoCo has negative pairs and BYOL does not, and this difference results in different optimal cosine similarity thresholds. We varied the cosine similarity in the range of 0.1-0.9 with a step size of 0.1. In Fig. 4, we show the training curves for contrastive learning with different cosine similarity threshold for both MoCo and BYOL. The training loss of MoCo increases as we increase the cosine similarity threshold. Correspondingly, cosine similarity threshold of 0.9 introduces the most similar negative pairs to the contrastive learning framework. The training loss of BYOL with cosine similarity threshold of 0.1 - 0.5 converged to a higher loss compared to cosine similarity threshold of 0.6 - 0.9. And among 0.6 - 0.9, threshold 0.7 has the lowest training loss.

An independent evaluation set (outside of all other training, validation, and testing sets) consisting of 2D RECIST slices and manual delineations for 38 necrotic lung lesions was used to optimize the cosine similarity threshold. As shown in Table 1, although cosine similarity of 0.9 in MoCo had the highest training loss, it had the best DSC on the evaluation set, manifesting that having challenging negative pairs could benefit the contrastive learning of MoCo. Whereas BYOL didn't use negative pairs, and cosine similarity threshold of 0.7 had the best mean DSC and mean HD. For both BYOL and MoCo, the segmentation dropped significantly with cosine similarity threshold ≤ 0.4 .

C. ABLATION STUDY

We conducted ablation study to evaluate the effect of the two proposed augmentation methods and also compared with

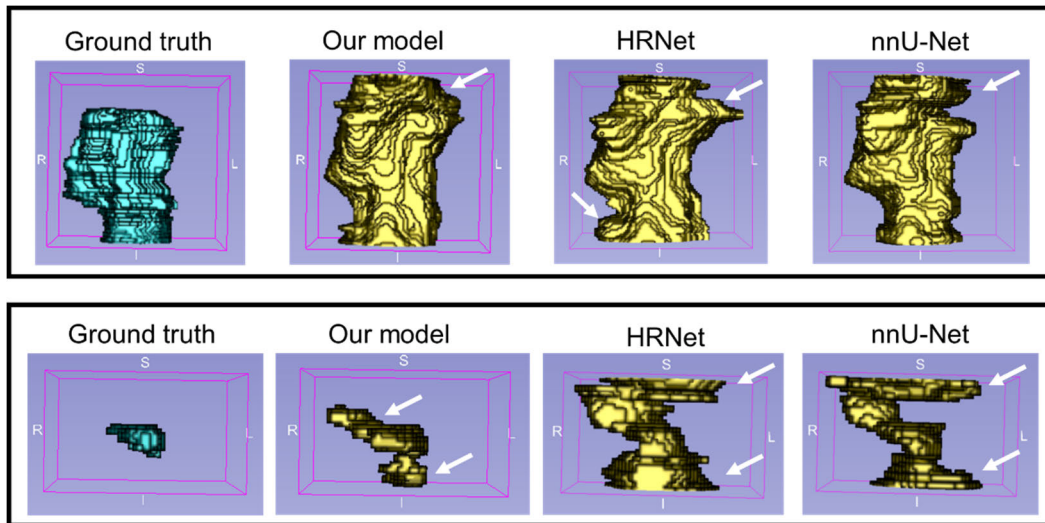


FIGURE 7. Volume rendering visualization of over-segmentation towards the superior and inferior ends of 3D lesion volumes. Ground truth rendering is shown in cyan, our MoCo pretrained model, HRNet, and nnU-Net models are rendered in yellow. The white arrows point to the over segmented regions.

training from scratch (Baseline) and tuning with ImageNet pretrained model (ImageNet). The DSC and HD on the three testing sets are shown in Table 2. Our dependency augmentation and mask-out augmentation are both beneficial for the improvement of necrotic lesion segmentation. The best model is MoCo with both augmentations, bringing 8.42% and 2.40% DSC improvement in 3D necrotic testing set compared to Baseline training from scratch and ImageNet transfer learning, respectively. For regular and DeepLesion testing set, MoCo with two augmentations also performed the best. Overall, MoCo performed better than BYOL. The original BYOL model performed even worse than the Baseline model trained from scratch. With our augmentation methods, the performance of BYOL substantially improved on all testing sets, demonstrating the effectiveness of our augmentation methods in BYOL's contrastive learning framework with only positive pairs. Example necrotic and regular lesion segmentation results from Baseline model, ImageNet pretrained model, MoCo with two augmentations, and BYOL with two augmentations are shown in Fig. 5.

D. COMPARISON WITH STATE-OF-THE-ART MODELS

We compared two state-of-the-art lesion segmentation models, HRNet and nnU-Net, with our proposed MoCo with two augmentations. HRNet and nnU-Net were trained with the same supervised training data and we repeated the training 3 times to evaluate consistency at model level. Results are shown in Table 3. Our proposed model has the highest DSC on all three test sets, and the DSC on the necrotic lung lesions were higher than those of the other two models by a significant margin.

E. LABELED DATA EFFICIENCY STUDY

We varied the amount of labeled data used in supervised fine-tuning at 1%, 10%, 20%, 40%, 60%, 80%, and 100%,

and show the DSC results on the 3D necrotic testing set for Baseline model, ImageNet pretrained model, MoCo with two augmentations, and BYOL with two augmentations in Fig. 6. MoCo with two augmentations and 10% labels reached better performance than Baseline model trained with all labels. Similarly, MoCo with two augmentations and 20% labels reached comparable performance as ImageNet pretrained model tuned with all labels. BYOL with two augmentations shared the same trend as MoCo but performed slightly worse than MoCo consistently.

V. DISCUSSION

The proposed MoCo model with two novel augmentations outperformed other models with pretraining and trained from scratch. Cosine similarity of 0.9 in MoCo introduced challenging negative pairs and worked the best in practice; cosine similarity of 0.7 in BYOL introduced the optimal amount of variation in positive pairs compared to other cosine similarity thresholds. Without negative pairs, strong augmentations to introduce more variations in positive pairs could be important for the success of BYOL.

Since our data for supervised tuning are the labeled RECIST slices, while the testing images are from the entire 3D lesion volume, we need the model to not only learn from the labeled RECIST images, but also adapt to the adjacent slices where lesions appear to be smaller. Pretraining of SiBA-Net model with our proposed augmentation methods allows better learning of a lesion's dependent features in z-dimension and in longitudinal scans, making the model more adaptive to the lesion size change between slices. The RECIST-based mask-out enables stable segmentation performance with the presence of a necrotic core, making the model more robust. In addition, our proposed SiBA-Net model, HRNet, and nnU-Net have 1,815,264, 65,846,401,

and 7,930,849 parameters, respectively. Thus, HRNet and nnU-Net are easier to overfit than the proposed model, resulting in over segmentation in the superior and inferior ends of 3D lesion volumes as shown in Fig. 7.

VI. CONCLUSION

In this work, we proposed two effective augmentation methods, dependency augmentation to capture inter-slice dependency, and distance transform-based mask-out, under the contrastive learning framework to improve the segmentation performance on necrotic lung lesions in CT images without sacrificing performance on regular lung lesions. Experiments demonstrated that the proposed MoCo method with the two augmentations led to the best performance on 3 test datasets including one dedicated to necrotic lung lesions.

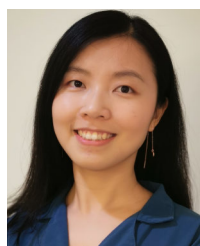
Our study has some limitations. Firstly, our pretrained model also has some over-segmentation towards the superior and inferior ends of 3D lesion volumes since only 2D annotated RECIST slices were used for fine-tuning. Using 3D delineated lesions for supervised fine-tuning in the future may resolve this issue. Secondly, the online calculation of cosine similarity between two images takes additional computation time. As a result, pretraining our proposed MoCo model takes about $\times 2$ the time of the original MoCo model, but this is a one-time effort. In the future, other similarity measurements such as mutual information and structural similarity index measure (SSIM) may be applied instead of cosine similarity. But a fast and PyTorch-based implementation is necessary. Thirdly, due to clinical demand, we only looked at necrotic lung lesions. In the future, we plan to extend the approach into other under-represented lesion subtypes in other organs.

With our proposed approach for improved segmentation of the necrotic lung lesions, we are able to measure the change of features such as necrotic volume and the percentage of necrotic region in the entire lesion volume. These features could be used in combination with the regular radiomics features for the prediction of the patient's response to treatments.

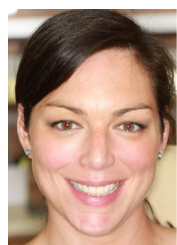
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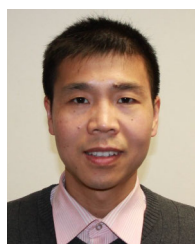
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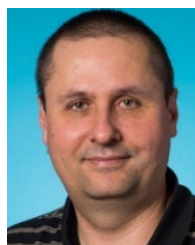
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